

# HAT-P-19b

R-1550

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_HATP-19\_220822 · created 2026-08-01T19:57:05+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2459813.8751 +/- 0.0045 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.165 +/- 0.053                 |
| Transit depth (Rp/Rs)^2                           | 2.72 +/- 1.74 [%]               |
| Orbital Inclination (inc)                         | 86.15 +/- 1.6                   |
| Airmass coefficient 1 (a1)                        | 0.97 +/- 0.29                   |
| Airmass coefficient 2 (a2)                        | -0.03 +/- 0.42                  |
| Scatter in the residuals of the lightcurve fit is | 28.24 %                         |
| Best Comparison Star                              | #1 - [441, 476]                 |
| Optimal Aperture                                  | 4.08                            |
| Optimal Annulus                                   | 8.48                            |
| Transit Duration (day)                            | 0.061 +/- 0.046                 |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.165 ± 0.053 vs 0.1418 (deviation 0.44σ) |
| Duration fit vs published | 0.061 d vs 0.1196 d (deviation 1.27σ)     |
| Mid-time O–C              | -9.7 min                                  |
| Depth SNR                 | 3.1                                       |
| Residual scatter          | 28.24 %                                   |

## Light curve

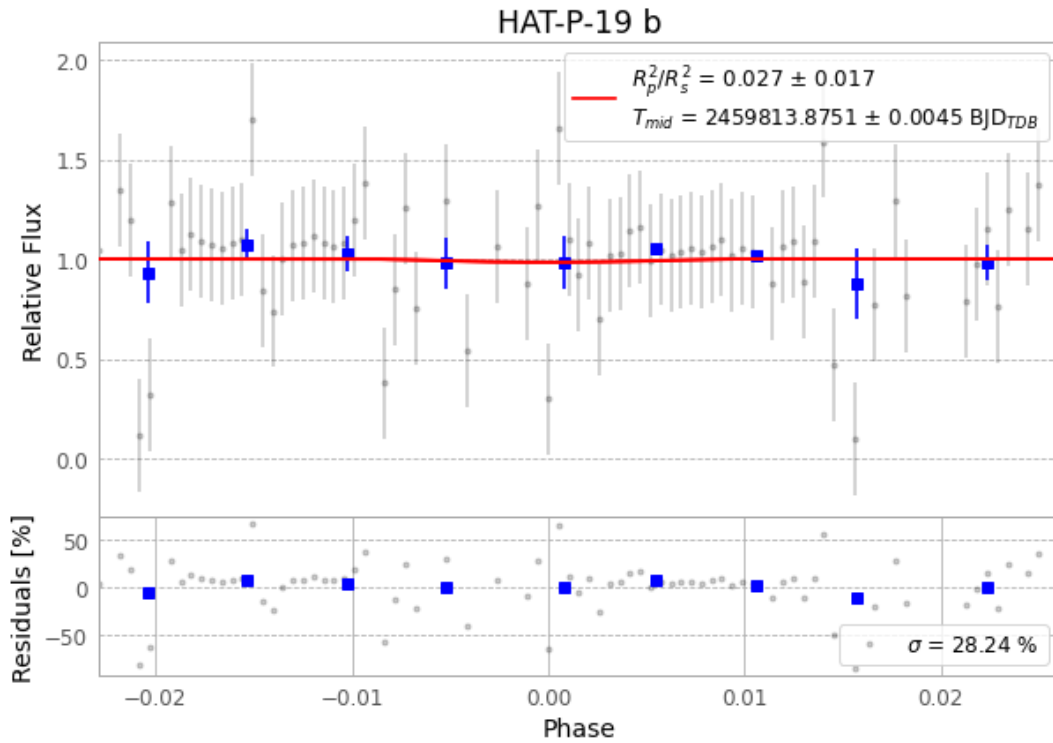


Figure 1 — FinalLightCurve\_HAT-P-19 b\_2022-08-21.png (SHA-256 e8353dcddb1f3dd3...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [512, 367], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 bfae75a626b0c417...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

94 FITS frames (62 MB), HATP-19220822064210.FITS → HATP-19220822112413.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-220822.log (9e5fa1b4c02c...), FinalLightCurve\_HAT-P-19 b\_2022-08-21.png (e8353dcddb1f...), FinalLightCurve\_HAT-P-19 b\_2022-08-21.csv (288181b584bc...)

## Compute resources

Wall time 230.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 19:57Z.